Is mass a contained rhythm, and time the same rhythm read? Eleven registered translations confronted with MICROSCOPE, lunar laser ranging, interferometry and the Margolus–Levitin bound

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

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Student version. This PDF is the student version of the record, written with an AI assistant from the archived text. The archived text remains on Zenodo: https://doi.org/10.5281/zenodo.23643718

Abstract

Physics already ties mass to time: every mass carries a rhythm, the Compton frequency mc^2/h, and all bodies fall in the same way within a few parts in 10^{15}. This deposit tests one thesis of the framework: the same counting is time when it is read, and mass when it is contained directly in the matrix of the point of observation . Read: it becomes a row of distinct events, like the ticks of a clock. Contained, in the reading chosen by the AI for the calculations: it stays inside the thing as a rhythm that produces no distinct events, like a top spinning in a closed box; in this reading mass is made of time, time held inside the thing. In the parametrisations compared, the data exclude that the count of the pieces weighs and are compatible with the value , the total energy including binding. It is the twin of deposit 2.62 on observed objects (DOI 10.5281/zenodo.23609967 , corrected in version 2.62.1): in the counting, fractions give objects and distances, multiples give time and mass.

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Student version of MM record 2.64.1 (DOI 10.5281/zenodo.23643718), written with an AI assistant from the archived text; the archived text remains on Zenodo.

Ugo Lanciano, October 2026. Licence CC BY 4.0 for the texts, MIT for the code.

What changes in version 2.64.1

Version 2.64.1 (25 October 2026): corrections and two new sections.

If you are reading this text for the first time, you can skip this list. You can come back to it after section 2, when the words will be clear.

The author asked for an independent check. The check found some places in version 2.64 that needed correcting. In those places the text presented as a result about the thesis something that depended on a choice. It was a translation choice made by the AI, and it was not declared. To translate, here, means to turn a sentence of the author into a precise definition, with which a calculation can be done.

This version corrects those places. In addition, it adds two new sections:

- section 9, on time with a direction;
- section 10, on mass in the matrix of whoever observes.

The programs, the calculations and the numbers of 2.64 do not change. Here are the eight points.

1. **Where the contained multiple is.** The author speaks of a multiple «contenuto direttamente nella matrice del punto di osservazione» ("contained directly in the matrix of the point of observation"). That is, inside whoever observes. Version 2.64 had read it in another way: as a rhythm closed inside the object. So all eleven translations are properties of the observed body. This is a choice of the AI. The results hold for these readings. The translation that uses the matrix of whoever observes is item M-i, in the new section 10. This translation coincides with standard physics.
2. **Count and value.** The text distinguishes the count of the pieces of a body from its value. This distinction was a reading by the AI of a premise of the author. The two ideas were written as precise formulas, called parametrisations. In those compared, the data exclude the count. And they are compatible with the value. But the premise, on its own, does not choose between the two readings.
3. **Masses that grow.** Item M-h says that masses grow with the age of the cosmos. This item is the AI's. The idea of starting the count at the beginning of the cosmos is also the AI's.
4. **The iron nucleus.** The author has a sentence on the average GCD. Using it to read the iron nucleus is a choice of the AI. The condition on the average GCD is not tested.
5. **The copies of the Compton rhythm.** Perhaps the Compton rhythm leaves copies: this is item M-g. That these copies would remain an internal record of the particle was a hypothesis of the AI. It is not tested. The separation from the observed objects holds in the static model of 2.62.
6. **The copies emitted by a source.** A source of light, like a star, emits copies. The number of these copies is a datum distinct from the mass. It is not derived from the mass. It depends on the conditions of the source, among them the illumination and the temperature.
7. **New section 9, "Physical time and time with a direction".** Here the direction of time is read as an increase of the relations in the observer's record. Read in this way, it is a quantity different from the direction of heat. At some moments they agree, at others they are opposite. No new prediction comes out of it.
8. **New section 10, "Mass in the matrix of whoever observes".** It is the translation that was missing: item M-i. Here mass is read as the part of the observer's copies that stays the same at every tick. Read in this way, mass coincides with standard physics. In no case examined does it give a different number.

Where to find the details of the corrections is written at the end, in the section *How the work was done*.

This work has a twin: deposit 2.62. A deposit is a work archived on Zenodo, a public archive for research. The two deposits are the two halves of a single step of the author's framework. It is the step farthest from the usual way of thinking about physics. In the counting, fractions give the observed objects and their distances: the deposit on objects, 2.62, deals with them. Multiples give time and mass: this deposit deals with them. Each of the two can be read on its own.

In short

The idea to be tested

This work puts a precise idea to the test: **mass and time are not two separate things, but the same rhythm in two different conditions.**

A rhythm is something that repeats, like a heartbeat or the tick-tock of a clock.

Two things physics already knows

Physics already knows two fundamental facts about mass.

- **First: every mass has a rhythm.** Every mass carries with it an extremely fast internal rhythm. It is called the Compton frequency and it is written mc^2/h. In words: the mass m, multiplied twice by the speed of light c, and divided by Planck's constant h. For an electron it is about $1.24 \cdot 10^{20}$ cycles per second. That is more than a hundred billion billion cycles in one second.
- **Second: all bodies fall in the same way.** In a gravitational field all bodies fall in the same way, whatever they are made of. This has been verified with great precision: down to a few parts in 10^{15}, that is, in a million billion.

The author's thesis

The author's thesis joins these two facts: **the same counting is time when it is read, and mass when it is contained**

directly in the matrix of the point of observation.

The words read, contained and matrix have a precise sense here. We make them clear in section 2. In this work "contained" was read as a rhythm closed inside the object. This is a choice of the AI, not the only possible one.

- **Read (it is time).** The rhythm comes outside. It becomes a row of distinct events, one after another. They are like the ticks of a clock, which mark the passing of time.
- **Contained (it is mass), in the AI's reading.** The same rhythm stays closed inside the object. It produces no events visible from outside. It is like a top spinning closed in a box, without leaving marks. When the rhythm stays closed in this way, it shows itself as mass.

In this reading, **mass is made of time: it is time held inside the thing.**

Count or value?

A body is made of pieces. What makes it weigh? Here two ideas are compared.

- The **count**: the sum of the single pieces that make up the body.
- The **value**: the total energy enclosed in the body. It also includes the binding energy, that is, the energy that holds the pieces together.

The two ideas were written as precise formulas, the parametrisations. In those compared, the data exclude that what produces weight is the count. And they are compatible with the overall value.

How we tested it

To test the thesis, we translated it in eleven different ways. Each translation is a mathematical definition, with which a calculation can be done. We registered all the translations before doing the calculations. To register means to write in a file and note its time and a check code. The code changes if even a single letter changes in the file. In this way one can see that the file was not touched afterwards. Together with the translations we registered a reference, that is, standard physics. And a threshold for neutrinos. Then we compared the translations with the measurements.

The four main results

- **The value weighs, not the count (in the parametrisations compared).** Suppose that what weighs were the count of the pieces. Then titanium and platinum would fall in a slightly different way. The difference would be 7.5 parts in 10,000. MICROSCOPE is a satellite that compared in orbit the fall of two cylinders, one of titanium and one of platinum. It sees no difference beyond a few parts in 10^{15}.
- **A mass at rest produces no distinct events.** Its state changes only by a phase. The phase is like the position of a turning hand: from outside nothing observable changes. If the same energy is released, it can produce at most two distinct states per Compton cycle. A theorem of Margolus and Levitin (1998) says so.
- **Whether the Compton rhythm leaves readable copies remains open.** A readable copy would be a mark that someone could read. The data we have in hand are not enough to decide.
- **Today no translation gives a prediction that differs from standard physics and can be tested.** The translations that hold coincide with $E = mc^2$ and with relativity. Only one translation departs from them: the checkerboard of deposit 2.40. But it gives an effect of 10^{-42}. It is a very small number. It is obtained by dividing one by a number written as a one followed by forty-two zeros. It is beyond the reach of any planned experiment.

All eleven translations read "contained" as a property of the body. The reading with the matrix of whoever observes is item M-i. It is in section 10, new in this version. There mass is the part of the observer's copies that stays the same at every tick. This reading too coincides with standard physics.

Physical time and time with a direction (section 9, new in this version)

Time has a direction: we remember yesterday, not tomorrow. The author's framework ties this direction to a growth. What grows is the number of relations, linked in a network, inside the matrix of whoever observes.

Read in this way, the direction of time and the direction of heat are two distinct quantities. Standard quantum models, reconstructed from published experiments, show it. In these models one looks at the relations in the observer's record. A record is a memory: the place where the marks of what has been observed remain.

- At some moments the relations are above their initial value, while heat goes from cold to hot.

- At other moments they are below it, while heat goes in the normal direction, from hot to cold.

The relations between the observed parts, instead, follow the heat. A known identity says so, that is, a mathematical rule that always holds. No new prediction.

How to read this text

- Section 2 presents the thesis without formulas. It is best read first. §2.3 shows it with three everyday examples. The sign § means section.
- Section 1 summarises what physics already knows. On a first reading it can be skimmed.
- Section 3 says how the thesis was tested.
- Section 4 gives the results. It starts with three calculations that can be done by hand.
- Sections 9 and 10 are new in this version. They deal with time with a direction and with mass in the matrix of whoever observes. They can be read on their own.

Technical words are made clear the first time they appear, and again in the glossary at the end. Numbers such as 2.25, 2.36.2, 2.38, 2.40 and 2.62 refer to earlier works of the author, deposited on Zenodo. The list with the DOIs is at the end of the references. A DOI is a code that points to a document forever, like the code of a book.

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1. What is already known

This section collects what physics already knows. It does not yet contain anything of the author's framework.

Mass and energy

Take a body at rest. Its mass is its total energy divided by c^2. Here c is the speed of light, and c^2 means c multiplied by itself. It is the formula $E = mc^2$, read backwards.

All the energies of the body count:

- the energy of the pieces;
- the energy with which the pieces move inside the body;
- the energy that holds the pieces bound, which is negative.

This is why a nucleus weighs less than its separate pieces. An example is iron-56. It has 26 protons, with their electrons, and 30 neutrons. Taken one by one, these pieces weigh 0.945 % more than the iron atom. The datum comes from the AME2020 tables, which collect the measured masses of atoms.

For the same reason, a box of mirrors with one joule of light inside weighs more than the empty box. It weighs $1.1 \cdot 10^{-17}$ kg more. A joule is about the energy needed to lift an object of a hundred grams by one metre.

Mass that comes from energy

Everyday matter is made of atoms. More than 99 % of its mass is in protons and neutrons. Protons and neutrons are made of smaller particles, the quarks, held together by gluons. About 95 % of the mass of protons and neutrons comes from the energy of the quarks and gluons enclosed in them. It does not come from the masses of the quarks.

Dürr and others (2008) recovered this with a lattice calculation. In this kind of calculation space and time are divided into many points, like the knots of a net. The calculation recovers the mass of the nucleon. Nucleon is the common name of proton and neutron. The calculation gives 0.936 GeV, with an uncertainty of about 0.03 GeV. The measured value is 0.939 GeV. The GeV is a unit of energy; for particles it is also used as a unit of mass.

The calculation starts from only three measured masses:

- those of the pion and the kaon, which fix the masses of the light quarks and of the strange quark;
- that of the Ξ particle (read xi), which fixes the scale.

If the scale is fixed with another particle, the Ω (omega), one gets 0.953 GeV.

The Compton rhythm

To every mass corresponds a frequency, that is, a number of cycles per second. It is written $\nu = mc^2/h$. The Greek letter ν is read nu. For the electron it is $1.24 \cdot 10^{20}$ cycles per second. For the neutron it is $2.27 \cdot 10^{23}$. Lan and others (2013) used this frequency to build a clock. It is a clock tied to the mass of an atom.

Can this rhythm be "seen" on its own? It is a debated question. Wolf and others (2011) studied atom interferometers. An interferometer splits a particle over two paths and then joins them again. Wolf and others show that the difference between the "Compton phases" of the two paths is zero. This holds in general relativity and in a wide class of theories. They also show that those experiments measure the fall, not the rhythm.

All bodies fall in the same way

All bodies fall in the same way: this is the equivalence principle. Here are the measurements.

- **MICROSCOPE.** The most precise measurement is that of the MICROSCOPE satellite (Touboul and others 2022). It compared an alloy of titanium and an alloy of platinum. It measured the relative difference of their accelerations. This difference is denoted by the Greek letter η, which is read eta. The result is $\eta = (-1.5 \pm 2.3 \pm 1.5) \cdot 10^{-15}$. The numbers after the $\pm$ sign are the uncertainties. The result is compatible with zero, that is, with no difference.
- **Lunar laser ranging.** Mirrors were left on the Moon. From the Earth laser pulses are sent, which bounce and come back. In this way the distance of the Moon is measured to the millimetre. Lunar laser ranging says two things. First: the Earth and the Moon fall towards the Sun in the same way, within $(-2.1 \pm 2.4) \cdot 10^{-14}$. Second: the mass that attracts and the mass that is attracted coincide, for aluminium and iron, within $3.9 \cdot 10^{-14}$ (Singh and others 2023).
- **Atoms in superposition.** A superposition is a state that combines two conditions, each with its own amplitude (a number from which the probability comes); a measurement finds only one of them. Atoms placed in a superposition of two internal states fall like the others. This holds within a few parts in 10^9 (Rosi and others 2017).
- **The future.** Projects under study aim at 10^{-17}.

Gravity and clocks

Near a mass clocks run slower. They all slow down in the same proportion. This was seen in two ways.

- A hydrogen clock, called a maser, was carried to an altitude of 10,000 km. The agreement with the prediction is within one part in 10^4.
- Two optical clocks were placed 33 cm apart in height, that is, one a little higher than the other.

Mass, time and distinct states

Two states are distinct when a measurement can separate them without error. They are like heads and tails of a coin. The technical name is orthogonal: perfectly distinguishable.

Margolus and Levitin (1998) proved a theorem. Take a system with mean energy E. The energy is counted from its ground state, that is, from the state of lowest energy. This system cannot pass through distinct states faster than a certain rate. In a long sequence it makes at most $2E/h$ passages per second. In words: twice the energy, divided by Planck's constant.

Lloyd (2000) turned this into a limit on computation. Imagine that all the energy mc^2 of one kilogram of matter is available. Then that kilogram could perform at most $5.43 \cdot 10^{50}$ logical operations per second. A logical operation is an elementary step of computation, like those of a computer.

The decisive point is where energy is counted from: its zero. A system can be in a state of definite energy. This state is called an eigenstate. A system in an eigenstate passes through no distinct state. Its state changes only by a phase. It is like a hand that turns while nothing observable changes.

Clocks and interference

A particle can go along two paths that then join again. Then light and dark stripes appear on a screen. They are called interference fringes. They recall water waves that pass through two openings and add up.

Proper time is the time marked by a clock that travels with the particle. Suppose that a particle carries with it a true internal clock. For example, two internal states that oscillate. And suppose that it goes along two paths with different proper times. Then the clock records which path it took. And the interference fringes lose contrast, that is, they become less sharp (Zych and others 2011).

Margalit and others (2015) showed this with a clock made of two spin states. Spin is a quantum property of particles that resembles a small rotation on themselves, but it is not a spinning ball. Margalit and others also recall a fact. In ordinary interferometry, without an internal clock, "time cannot yield 'which path' information".

A body made of many oscillating parts is different. For it the difference in proper time is enough to make interference disappear. Take one gram at room temperature, placed in two positions one micron apart, that is, a thousandth of a millimetre. Interference disappears in about a thousandth of a second (Pikovski and others 2015).

Feynman's checkerboard

A particle can be described as a path on a checkerboard. The path moves forward in a zigzag, along the directions of light rays. Mass is how often the path turns. This construction is called the Dirac quantum walk (Bialynicki-Birula 1994; Meyer 1996; Arrighi, Nesme and Forets 2014; D'Ariano and Perinotti 2014). It was put to the test in deposit 2.40.

Neutrinos

Neutrinos are extremely light particles. There are three of them, with three different masses. While they travel, neutrinos change type: these are the oscillations. From the oscillations the differences between the masses are obtained.

JUNO is a large neutrino detector. In 2026 it measured the differences between the squares of the neutrino masses, to one per cent. JUNO favours the normal ordering, the one in which the lightest is the first. It favours it at at least 2.1σ. The Greek letter σ (sigma) denotes the standard deviation, that is, the measure of the uncertainty. More sigma means a more certain result.

DESI is a large census of galaxies. DESI, together with the cosmic background and within the standard cosmological model, limits the sum of the neutrino masses. The sum is less than 0.064 eV. The cosmic background is the oldest light in the universe. The eV, or electronvolt, is a very small unit of energy. There is also a method that takes into account a fact: masses cannot be negative. With this method the limit drops to 0.053 eV. It is a value even lower than the minimum imposed by the oscillations, which is 0.0587 eV. The DESI collaboration itself points this out.

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2. The thesis and its reasons

2.1 The framework, in a few lines

In the author's framework the local world arises from a relation. The relation is counted through a common unit. This unit is the greatest common divisor of the relation, in short GCD. At school the GCD is used with numbers. Here the idea is the same.

An example with small numbers. We want to compare two quantities, 3 and 5. We need their largest common measure. Here it is 1: it is their greatest common divisor. To measure both quantities, this unit must be repeated three times in the first and five times in the second. In this way counting produces **copies** of the unit.

Whoever is inside the world, like us, arranges these copies in two ways:

- **in a row**, one after another: these are the **multiples**, and they appear as time;
- **side by side**, all together: these are the **fractions**, and they appear as space.

An image can help. Think of many identical little bricks. Placed one after another, they make a row that grows longer, like time. Placed one next to another, they cover a surface, like space.

The set of the copies is called the **matrix**. The point of observation of whoever looks is a part of this matrix.

2.2 What the thesis says

In the author's words: «Lo stesso multiplo, quando è contenuto direttamente nella matrice del punto di osservazione, si trasforma in massa, cioè in materia concreta, come la intendiamo di solito.» ("The same multiple, when it is contained directly in the matrix of the point of observation, turns into mass, that is, into concrete matter as we usually understand it.")

The author speaks of a multiple contained in the matrix of the point of observation, that is, in whoever observes. In this work we read it in another way, by a choice of the AI. We read it as a counting in a row that stays closed inside an object, without showing itself outside. In this reading concrete matter is made of contained time. All the translations of section 3 are properties of the body. The translation that uses the matrix of whoever observes is item M-i, in section 10.

Here is the thesis said in the words of physics, in the AI's reading.

- **The read multiple is time.** It is a row of distinct states, one after another. An observer can count them. They are the ticks of a clock.
- **The contained multiple is mass.** It is the same counting when it stays inside the thing. It is a rhythm that produces no distinct states. It changes the thing only by a phase. In physics this rhythm has a name: the Compton frequency.
- **What weighs is the value of the rhythm, not the count of the pieces.** The value is the total energy of the thing, binding included. The count of the pieces is the sum of the masses of the pieces taken one by one, without the binding.

Four words, defined at once:

- **multiple:** counting forward, one copy after another;
- **read:** becoming a row of distinct states, observable as time passing;
- **contained:** in the AI's reading, remaining an internal rhythm, a phase, without producing distinct states. For the author it means contained directly in the matrix of the point of observation;
- **value:** the total energy contained. Relativity calls it the rest mass multiplied by c^2.

2.3 Three everyday examples

The closed top. A top spins inside a sealed box. From outside one cannot count how many turns it makes: it leaves no marks. Yet the box weighs more. For 100 joules of rotational energy it weighs $1.1 \cdot 10^{-15}$ kg more. This is known physics: it is $E = mc^2$. If the top hits the wall, its turns become events: a noise, a little heat. Contained, the rhythm weighs. Read, it becomes events in time.

Light in a box. A ray of free light has no mass. It has the frequency of its wave. But it has no rhythm at rest, that is, no Compton frequency. Along its path proper time is zero. And light always travels at the speed of light. Now let us close the same light in a box of mirrors. The light gives the box a mass: for one joule of light, $1.1 \cdot 10^{-17}$ kg. The same energy, contained, becomes mass.

The iron nucleus. The pieces of iron-56, separated, weigh 0.945 % more than the nucleus. When the pieces join, part of their value leaves as radiation. It becomes read, that is, it becomes events that flow in time. So the mass that stays contained decreases. In the reading of the thesis: less rhythm closed inside, more events released outside.

A sentence of the author goes in the same direction. It says that if, coming closer, the average GCD decreases, the total energy decreases too. The exact words are in a separate file, the background file. Reading the iron nucleus with this sentence is a choice of the AI. The author's condition on the average GCD has not been tested.

What weighs is what remains: the value, not the number of pieces.

2.4 Why think of it this way

1. **Physics already ties mass and time together.** A mass is a rhythm: the Compton rhythm. Near a mass all rhythms slow down together. The thesis does not add a new link. It says that mass and time are the same counting in two conditions.
2. **A single rule separates the two things:** read or contained. The rule introduces no new constant. A constant is a fixed number of nature, like the speed of light. The units of measurement pass through h and c, as in known physics.
3. **The bridge between the two conditions is a known theorem.** The Margolus–Levitin bound says how many distinct states per second a given energy can produce, at most. Read with the thesis, the contained multiple is a

store of possible time. It becomes time when it is released. It is the "mass as frozen time" of deposit 2.25.

4. **Consistency with the starting framework.** In the framework the multiple is mass. And mass and time are two faces of the same counting. The author's words are in the background file, kept apart from this text.

2.5 How the thesis reads some known facts

- **$E = mc^2$.** The value of the contained rhythm is the energy. It is the same equality, read as "what is contained weighs".
- **The mass defect.** Binding makes part of the value leave as radiation. So what stays contained is worth less than the sum of the pieces. It is the case of iron (§2.3).
- **Massless light and the box of light.** What has no rhythm at rest, that is, no Compton frequency, has no mass. The same energy closed in a box does (§2.3). The author's book says it like this: what has no weight, like light, is «ciò che non batte» ("what does not beat").
- **The proton mass.** About 95 % is energy of the quarks and gluons closed inside. It is contained energy, hence mass.
- **Universal free fall.** Suppose that what weighs is the value, that is, all the contained energy. Then every body falls in the same way, whatever its composition. This is the equivalence principle. The author's book says that near a mass the shift of the rhythms «rigradua insieme, e nella stessa proporzione, tutti i ritmi leggibili da quel punto» ("rescales together, in the same proportion, all the rhythms readable from that point").
- **Slower clocks near a mass.** Let us stand far from a large mass and look near it. From there we see two things. The time of clocks, that is, the read rhythm, slows down. And the energy of a body, that is, the contained rhythm, is worth less. The two things change exactly in the same proportion. On the spot, instead, the measured mass of that body does not change. In the reading of the thesis, gravity, mass and time move together because they are the same rhythm. In the AI's reading this is also the author's sentence of §2.3, read for gravity: if, coming closer, the average GCD decreases, the energy decreases. The condition on the average GCD is not tested.

2.6 What it costs, said plainly

- **Mass is not a substance.** A mass at rest does not "do" anything distinguishable: its state changes only by a phase. It looks like a paradox. But it is what quantum mechanics says of every state of definite energy.
- **The value of a single mass is a datum.** The thesis does not say how much the electron or the proton weighs. It takes it from measurements, as standard physics does. In the author's framework the value of each mass depends on how the matrix is built. And it cannot be derived (background file).
- **Today it gives no new numbers.** The translations that hold coincide with standard physics (section 4).
- **One question remains open:** whether the Compton rhythm leaves readable copies (§4.4).

2.7 What is new with respect to the closest studies

We did not carry out a systematic search of the literature, that is, of all the articles on the topic. The closest works we know are these.

- **Margolus and Levitin (1998), Lloyd (2000).** They tie energy, and hence mass, to the maximum number of distinct states per second. Here that link becomes the bridge between the two conditions of counting: read and contained.
- **Zych and others (2011).** They distinguish two things: the phase accumulated by a particle and the internal clock that records the path. And they propose to test a hypothesis: that proper time is a quantum degree of freedom. A degree of freedom is a way in which a system can change. Here the same distinction separates the translations. On one side, those in which the Compton rhythm remains a phase. On the other, the one in which it would leave copies.
- **Wolf and others (2011)** discuss whether the Compton rhythm is observed in interferometers. **Lan and others (2013)** make a clock of it. **Greenberger (2010)** discusses the conceptual problems that tie time and mass in quantum mechanics.
- **Pikovski and others (2015), Zych and Brukner (2018).** They treat the mass of a composite body as its internal energy, in relativity and in quantum mechanics. It is the same thing that here is called the value.
- **Dirac quantum walks.** They give mass as a turning amplitude on a lattice. Deposit 2.40 is a case of it.

The step that belongs to this work is another. One takes the thesis "the contained multiple is mass". One translates it in eleven ways, all as properties of the body. One registers the translations before the calculations. And one lets the data choose. The data exclude that the count of the pieces weighs. And they are compatible with the value. Another question remains: does the contained rhythm remain only a phase, or does it also leave readable copies? The data at hand do not decide it (§4.4).

3. How the thesis was tested

3.1 The thirteen items

Each way of translating the thesis is an item of a list. Each item has a short name, like M-a or M-b1. In the test of 2.64 there are thirteen items. Version 2.64.1 adds item M-i, in section 10.

The thirteen items are:

- eleven translations, that is, eleven mathematical definitions of "contained multiple = mass";
- the reference M-0, that is, standard physics;
- the threshold N for neutrinos.

All were written and registered before the calculations. Next to each one we wrote the outcome we expected.

Here is what each item says, in short.

- **M-0.** No consequence beyond standard physics. It is the reference.
- **M-a.** Mass is the rhythm of the multiple in proper time, that is, the Compton rhythm. It uses a postulate of deposit 2.25: the phase of the count is equal to the action divided by $\hbar$. A postulate is a rule taken as a starting point. The action is a quantity of physics that is measured in energy times time. $\hbar$ is read h-bar and is a form of Planck's constant.
- **M-b1.** Mass is the turning angle per step of the checkerboard of deposit 2.40. The walkers are nucleons and electrons. The step is the Planck time, an extremely small time scale built with h, c and G.
- **M-b2.** As M-b1, but a whole body is a single walker.
- **M-c.** Mass is how much the body slows the rhythms in the space around it. This slowing is summed over a surface that encloses the body. It is called the Komar mass.
- **M-d1.** Mass is the number of copies of the source. It is premise P4 of deposit 2.36.2. A premise is a starting sentence of the author. Here the copies are the pieces counted without binding.
- **M-d2.** As M-d1, but the copies carry the value: the total energy, binding included.
- **M-e.** Mass and the slowing of time. It is a premise of the author, in the background file. Seen from far away, the value of a mass near another drops. It drops in the same measure in which clocks slow down.
- **M-f1.** Mass is the number of distinct states per second. The energy is counted from the state at rest.
- **M-f2.** Mass is the ceiling, that is, the maximum limit, on the number of distinct states per second. The energy is counted from the vacuum.
- **M-g.** The Compton rhythm leaves a readable copy at every cycle.
- **M-h.** Mass is the number of multiples accumulated since the beginning of the cosmos. And it grows as the age of the cosmos. This item is the AI's. The choice of starting the count at the beginning of the cosmos is also the AI's.
- **N.** It is not a translation. It is the decision threshold for neutrinos of deposit 2.38.

3.2 The data

We used these measurements.

- **MICROSCOPE**, with titanium and platinum.
- **Lunar laser ranging.** It gives three things: the comparison between aluminium and iron; the fall of the Earth and the Moon; the constancy of G in time. G is the gravitational constant. For the variation of G lunar laser ranging gives $(-5.0 \pm 9.6)\cdot 10^{-15}$ per year.
- **A neutron interferometer**, with its two arms at different heights. An arm is one of the two paths. The arms are about one centimetre apart. The coherence time is about 10^{-4} s, that is, a ten-thousandth of a second. It is the time during which the two paths remain able to interfere. These values are those reported by Zych and others (2011).
- **The atomic masses** of the AME2020 tables.
- **The Margolus–Levitin and Lloyd bounds.**
- **JUNO 2026 and DESI**, for neutrinos.

The physical constants are those of the Particle Data Group (2025). It is the group that collects the reference values of particle physics.

Some values are known and were taken without downloading a source. We declare it. They are these:

- the natural isotopic abundances, that is, how much of each variety of an element there is in nature;
- the radius and the mass of the Earth and the Moon;
- the age of the cosmos, 13.8 billion years.

No outcome depends on these values beyond a few per cent.

3.3 The criteria, fixed in advance

There are seven criteria, the same for all items. Each item was checked on each of them.

1. $E = mc^2$ and the mass defect.
2. Universal free fall, within the precision of MICROSCOPE.
3. Mass that comes from energy: the nucleon at 95 %.
4. No new constant beyond h, c and G.
5. No mass value derived by fitting, that is, by choosing a number on purpose so that it matches the measurement.
6. Relativity: clocks in motion and clocks in gravity.
7. The limiting cases: no multiple, no mass; light without mass; a large mass fixed in the matrix.

The outcome rules were also fixed in advance. An outcome is the result of the comparison with the data. Here are the rules.

- **Excluded:** the prediction is more than three standard deviations from the datum, or exceeds the published limit.
- **Compatible:** the prediction is within two standard deviations of the datum.
- **Not testable today:** the predicted effect is at least ten times below today's sensitivity.
- **Distinctive:** the effect exceeds the sensitivity of an experiment existing or planned by 2035, without being excluded by the data. A distinctive item would say something different from standard physics, and it could be checked.
- **Not conforming:** a criterion cannot be checked, because the translation does not define something that is needed.

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4. What comes out

4.1 Three calculations that can be done by hand

First calculation: the count and the value. Take the two MICROSCOPE alloys. For each atom we take its mass. We compare it with the sum of the masses of its pieces, taken one by one.

- For the titanium alloy the count of the pieces exceeds the value by 0.9342 %.
- For the platinum alloy it exceeds it by 0.8585 %.

Suppose that what weighs were the count of the pieces. Then the two alloys would fall with a discrepancy, that is, with a small difference. The discrepancy would be about the difference between these two numbers: 7.5 parts in 10,000. MICROSCOPE instead measures a discrepancy compatible with zero. The error of the measurement is 2.75 parts in 10^{15}.

Second calculation: the copies of the Compton rhythm. In a neutron interferometer the two arms are at different heights. Whatever is higher has a proper time that flows a little faster. Take a height difference of one centimetre, for a time of 10^{-4} s. The difference in proper time between the two arms is $1.09 \cdot 10^{-22}$ s. The Compton period of the neutron, that is, the duration of one of its cycles, is $4.40 \cdot 10^{-24}$ s. That makes about 25 cycles of difference.

Suppose that each cycle left a readable copy. Then the two arms would carry records differing by 25 copies. The calculation is simple: $\Delta N = m\, g\, \Delta h\, t / h \approx 24.8$. In words: the mass of the neutron m, times the acceleration of gravity g, times the height difference Δh, times the time t, all divided by Planck's constant h. The result is about 24.8, that is, about 25.

Third calculation: one kilogram. The energy of one kilogram is $mc^2 \approx 9 \cdot 10^{16}$ J. Lloyd's bound gives at most $5.43 \cdot 10^{50}$ operations per second. That is four logical operations per Compton cycle. For a long sequence of distinct states, Margolus and Levitin give half of that: two states per cycle.

4.2 Count excluded, value compatible, in the parametrisations compared (M-d1, M-d2)

Item M-d1 says: "the copies are the pieces counted without binding". This translation predicts a difference in fall between titanium and platinum: $\eta = 7.5 \cdot 10^{-4}$. It is $2.7 \cdot 10^{11}$ times the error of MICROSCOPE: hundreds of billions of times.

One can also ask what part of gravity could act on the count of the pieces, instead of on the value. Let us call this part ε (epsilon). The data say $\varepsilon \leq 5.3 \cdot 10^{-12}$. That is, at most one part in $2 \cdot 10^{11}$ of gravity. Counting the nucleons, instead of the masses of the pieces, gives the same order of magnitude: $\eta = 8.0 \cdot 10^{-4}$. An order of magnitude is a factor of ten.

The same translation can also be applied to the mass that attracts, with aluminium against iron. It exceeds the lunar-laser limit by $1.3 \cdot 10^{10}$ times.

It can also be applied to the gravitational binding of the Earth and the Moon. The binding is estimated by thinking of the Earth and the Moon as uniform spheres, that is, with the same density everywhere. For the Earth the binding is $4.2 \cdot 10^{-10}$ of its energy. For the Moon it is $1.9 \cdot 10^{-11}$. The translation then predicts that the Earth and the Moon fall towards the Sun with a difference of $4.0 \cdot 10^{-10}$. The measurement gives $(-2.1 \pm 2.4) \cdot 10^{-14}$. The distance is $1.7 \cdot 10^{4}$ standard deviations, that is, about seventeen thousand.

Be careful about what this means. These are exclusions of a parametrisation within the models with which those measurements are read. They are not separate measurements of each mass.

Item M-d2 says: "the copies carry the value". It is compatible with everything: it is standard physics.

So, in the parametrisations compared, the data exclude that the sum of the masses of the pieces weighs. And they are compatible with the value, that is, the total energy, binding included. The reading that falls is M-d1, the reading of the count.

A correction on this point. There is a premise of the author that concerns the registers. It says that what passes from one register to another is only the partial premise of the new history. The new history is formed with the present register. The exact words are in the background file. In September 2026 we had read this premise as a distinction. On one side the count, which is transmitted. On the other the value, which is formed in the register that receives. That reading is the AI's. The premise on its own does not choose between count and value. And the data do not test it.

4.3 The contained multiple produces no distinct states (M-f1, M-f2)

Take a particle at rest, in a state of definite energy. We can compare the initial state with the state after any time. One measures how much the two states resemble each other. In technical terms it is the overlap between the two states: here the word means resemblance, not the state that combines two conditions. This overlap is always 1 in modulus, that is, the maximum. The state changes only by a phase.

Item M-f1 says: "mass is the number of distinct states per second". This translation would therefore give zero mass to an electron at rest. It falls by its own definition.

Item M-f2 holds instead: it is the reading as a ceiling. The energy is counted from the vacuum. The same energy mc^2, if it becomes evolution, allows at most two distinct states per Compton cycle, in a long sequence. In Lloyd's count these are four logical operations per cycle.

Together, the two items give the AI's reading in precise form: **the contained multiple, read as a rhythm inside the body, is mass and produces no distinct states; when it is released it becomes time, with a maximum rate fixed by the mass itself.**

4.4 Does the Compton rhythm leave copies? (M-g)

Suppose that at every Compton cycle matter left a readable copy. Let us go back to the neutron interferometer of §4.1. The two arms would carry records differing by about 25 copies. The records could be distinguished, and the fringes would disappear.

Fringes of neutron interferometers in gravity have been observed since 1975 (Colella, Overhauser and Werner). But this is not enough to close the question. The reasons are two.

- The measurement of the contrast of the fringes, in the geometry used, is not among the sources of this work.

- The sources do not agree on how to read the observed gravitational phase. For Zych and others, in their scheme, it is the phase difference due to proper time. For Wolf and others, in atom interferometers, the difference of the Compton phases between the paths is zero.

There is a nearby hypothesis: that proper time is a quantum degree of freedom. Zych and others present it as a possibility to be tested.

Outcome: not concluded.

One thing remains firm. In standard quantum mechanics a phase that changes in the same way across the whole state gives no information about the path (Margalit and others 2015). A neutron in its ground state has no internal degree of freedom that changes state at every Compton cycle. For a readable copy to remain, an extra degree of freedom would be needed. It would be as in the hypothesis of a quantum proper time of Zych and others. M-g, therefore, does not follow from standard physics. It remains not concluded, as a hypothesis that goes beyond standard physics.

To decide, two things are needed: the original 1975 source and an analysis of proper time in the two arms.

A correction. In a first draft of the work this outcome was: excluded. After the verification by a second artificial intelligence it was corrected to: not concluded (section 8).

4.5 The checkerboard of deposit 2.40 (M-b1, M-b2)

On the checkerboard mass is the turning angle per step. Deposit 2.40 has already shown that the fall depends slightly on mass. The formula is $a = -g\,\theta/\tan\theta$. In words: a is the acceleration of fall; g is the normal acceleration of gravity; θ (theta) is the turning angle; tan is the tangent, a function of angles studied in trigonometry. The formula gives a correction of about $\theta^2/3$, that is, the angle multiplied by itself and divided by three.

A real body is made of many walkers. For such a body two rules are needed that deposit 2.40 does not give. Here they are chosen and declared:

- the walkers are nucleons and electrons;
- the binding energy falls without correction, like a massless walker.

Take the step equal to the Planck time. Then titanium and platinum would fall with a difference of about $1.1\cdot10^{-42}$. With a three-dimensional estimate the difference is $4.5\cdot10^{-42}$. It is twenty-five orders of magnitude below the future projects, which aim at 10^{-17}. Twenty-five orders mean a factor written as a one followed by twenty-five zeros.

Outcome of M-b1: not testable today. And not conforming to one criterion. Deposit 2.40 does not show that massless walkers, bound together, form a body with mass.

And if the step were not the Planck one, but free? Then MICROSCOPE would require a step shorter than about 10^{-21} m. Said as an energy, the energy of the step would have to be above $1.5\cdot10^{5}$ GeV. With the three-dimensional estimate, above $3.1\cdot10^{5}$ GeV. These numbers use the widest end of the two-standard-deviation interval.

Item M-b2 treats a whole body as a single walker. It would have a turning angle of millions of radians. The radian is a unit for measuring angles: a full turn is a little more than six radians. But on the checkerboard an angle that exceeds a full turn (360 degrees) starts again from the beginning. It is like the hand of a clock: after one turn it comes back to the same point. So the checkerboard does not distinguish that body from a much lighter walker. And so it does not assign it a unique mass, that is, a single well-defined mass.

Outcome of M-b2: not conforming.

A correction. In a first draft the outcome of M-b2 was: excluded. After the verification it was corrected to: not conforming (section 8).

4.6 Masses that grow with the age of the cosmos (M-h)

Item M-h is the AI's. Starting the count at the beginning of the cosmos is also a choice of the AI.

Suppose that mass were the number of multiples accumulated since the beginning of the cosmos. Then mass would grow by one part in 13.8 billion per year, that is, by $7.2\cdot10^{-11}$ per year.

Lunar laser ranging measures a combination of quantities: $G\,m^2/\hbar c$. Let us keep G, h and c fixed. In this combination mass appears squared, that is, multiplied by itself. The combination would change by twice as much: $1.45 \cdot 10^{-10}$ per year. The measurement gives $(-5.0 \pm 9.6) \cdot 10^{-15}$. The distance is $1.5 \cdot 10^{4}$ standard deviations, that is, about fifteen thousand.

Outcome: excluded. The reading that falls is M-h, the AI's item on masses that grow. The outcome holds with the declared mapping. The mapping is the way in which the growth of masses was connected to what lunar laser ranging measures. The full passage to the lunar-laser parameters is not derived here.

4.7 The readings that coincide with standard physics (M-a, M-c, M-e)

Three items coincide with standard physics.

- **M-a:** mass as the Compton rhythm.
- **M-c:** mass as the slowing of the rhythms around the body. Here a word is needed: the metric. It is the rule that says how distances and times are measured in a certain space. The Komar mass of the metric of deposit 2.36.2 equals M at every radius, that is, at every distance from the centre.
- **M-e:** mass as a value that, seen from far away, drops together with the rate of clocks. With this item the Earth and the Moon fall in the same way, 0.88 standard deviations from the measurement.

They are identities or coincidences by construction. They are not independent checks of the thesis. But they show that the thesis is not contradicted at any of the points tested.

One thing about deposit 2.36.2 should be recalled. There the Schwarzschild metric comes from a declared postulate, the transfer postulate. It does not come from the counts alone. The Schwarzschild metric describes space and time around a mass shaped like a sphere, like a star.

4.8 All outcomes

The symbol σ denotes the standard deviations.

item	key number	outcome
M-0	—	reference
M-a	identity	compatible, equal to standard physics
M-b1	$\eta \approx -1.1 \cdot 10^{-42}$	not testable today; not conforming (the mass of a composite body is not shown)
M-b2	an angle that starts again after a full turn	not conforming
M-c	Komar mass = M at every radius	compatible, an identity of general relativity
M-d1	$2.7 \cdot 10^{11}$ σ; $1.3 \cdot 10^{10}$ times the limit; $1.7 \cdot 10^{4}$ σ	excluded
M-d2	—	compatible, equal to standard physics
M-e	0.88 σ	compatible, equal to standard physics
M-f1	no distinct state at rest	excluded by its own definition
M-f2	2 states per cycle (4 operations in Lloyd's count)	compatible
M-g	about 25 copies of difference	not concluded
M-h	$1.5 \cdot 10^{4}$ σ	excluded, with the declared mapping (an item of the AI)
N	$\Sigma = 0.0591$ eV	dated threshold registered

No item is distinctive. All read "contained" as a property of the body. Item M-i of section 10, which uses the matrix of whoever observes, is not distinctive either.

4.9 Neutrinos: a threshold with a date

Leptons are a family of particles: the electron and the neutrinos belong to it. Deposit 2.38 read the lepton masses with the Koide relation, a formula that ties three masses together. From it, it drew a prediction for the neutrinos. It is the same prediction as Brannen's (2006).

With the JUNO 2026 data the same relation gives three masses: 0.00036, 0.00860 and 0.05009 eV. The sum is 0.0591 eV. If the JUNO data are varied within their errors, the sum lies between 0.0588 and 0.0594 eV. It is 0.37 meV above the minimum of the normal ordering. A meV is a thousandth of an electronvolt.

The rules were registered on 24 October 2026, with a deadline of 31 December 2035. Here they are.

- The reading falls if the inverted ordering is established at at least three standard deviations.
- The reading falls if a measurement of the sum (not a limit) places it above 0.0593 eV by more than three standard deviations.
- Suppose that a measurement gave an upper limit at 99.7 %, that is, with a certainty of 99.7 %, below 0.0587 eV. It would exclude the whole normal ordering with the lightest neutrino almost massless. It would not exclude only this reading.
- A specific confirmation is not possible. The difference from the minimum is less than one thousandth of an electronvolt. It is beyond the reach of foreseeable cosmology.

Today the DESI limit with the Feldman–Cousins method is 0.053 eV. It lies below both the prediction and the minimum of the normal ordering. This tension is between cosmology and neutrino oscillations. It does not concern this reading alone.

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5. What is not claimed

This section says clearly what the work does not maintain.

- Mass is not derived, nor the value of any mass. The masses of the particles remain measured data.
- It is not claimed that the author's framework explains mass. What is claimed is this. The framework reads mass as the contained multiple. With the definition "the contained multiple is the value and remains a phase" the reading coincides with $E = mc^2$ and with relativity. The alternatives tested fall or remain open, as said in section 4.
- The compatible items coincide with standard physics by identity or by construction. They are not independent proofs.
- It is not claimed that mass is a rhythm closed inside the object. This is the AI's reading of the word "contained". The author speaks of a multiple contained in the matrix of the point of observation. The translation that uses the matrix of whoever observes (section 10) also coincides with standard physics.
- It is not claimed that the neutron fringes exclude the copies of the Compton rhythm (§4.4).
- It is not claimed that the checkerboard predicts a violation of the equivalence principle. With the Planck step the effect is not testable. And the mass of a composite body, on the checkerboard, is not shown.
- The neutrino prediction is not new. It is Brannen's (2006), updated with the JUNO data.
- Some things were not studied here:
- the fall on the checkerboard with copies arranged at random, as in causal sets (Johnston 2008; Dowker, Henson and Sorkin 2004);
- a rule that fixes the step of the checkerboard other than the Planck one;
- the mass threshold of the quantum world. This is treated in a separate work, with a prediction dated to 2035.

6. The contact with observed objects

The twin deposit, 2.62, deals with the observed objects. Take an object on which two observers agree. Deposit 2.62 reads it as the common part of their records. This common part is made of copies taken at the same moment. It is a static model: the copies are taken all together, and not followed in time.

In that static model the compatible translations of mass do not change the result. In them the contained multiple produces no distinct states. So it adds no copies to the observers' records. Whether item M-g, not concluded, could change this

result remains open.

If the open item M-g were true, the contained multiple would leave copies. In version 2.64 we had written something about those copies. We had written that they would be an internal record of the particle. And that they would not enter the observers' records without an interaction. But this is a hypothesis of the AI, not tested.

The separation from the observed objects holds in the static model of 2.62, with the copies taken at the same moment. Real copies, instead, accumulate in time. In deposit 2.62 the projection (or reading) is what the observer obtains from the record by bringing it to a simpler question. Whether the projection can be defined without the multiple remains open.

In deposit 2.62 distance is read by triangulation. To triangulate means to look at an object from two different points and measure the angles, as surveyors do. There is also another way: distance from luminosity, that is, from how bright an object appears. This distance needs the number of copies emitted by the source. That number depends on the conditions of the source, among them the illumination and the temperature. It is not derived from the mass.

For gravity the mass that counts is the value. Gravity around a body spreads according to Gauss's law: it depends on what is inside a closed surface around the body. What spreads in this way is the total energy (item M-d2).

7. How to redo the calculations

All the calculations can be redone with a few programs, on an ordinary computer. A few seconds are enough for the part on mass and about three minutes for section 9. The list of the programs, the files needed and the steps to follow are at the end, in the section *How the work was done*.

8. Where this work comes from

The author is not a physicist. The translations, the programs and the calculations were made with an artificial intelligence. A second artificial intelligence verified them, in two rounds.

Thanks to these verifications, two results were corrected:

- the copies of the Compton rhythm (item M-g): from excluded to not concluded;
- the whole body on the checkerboard (item M-b2): from excluded to not conforming.

In version 2.64.1 an independent check, requested by the author, found the undeclared translation choices. They are the ones corrected in this version, listed at the beginning. Each point was rechecked against the original files.

Who did the checks, how many rounds there were and where the reports are is written at the end, in the section *How the work was done*. The author's sentences from which the work arises are in the background file.

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9. Physical time and time with a direction

New section in version 2.64.1. It can be read on its own.

The previous sections treat time as the read multiple. It is a row of distinct states, like the ticks of a clock. It is **physical time**, the time that clocks measure.

A row of ticks, however, does not say by itself which way one is going. The laws of microscopic physics work almost in the same way forwards and backwards. Yet we remember the past and not the future. And heat passes from hot to cold: a cup of tea cools down, it does not warm up by itself. This direction is **time with a direction**.

The author's framework gives a reading of it. The direction of time is the increase in the number of relations, linked to one another in a network, inside the matrix of whoever observes. The exact words are in the background file.

This section says three things:

- what physics knows about the direction of time;
- how we translated the author's sentence;

- what comes out of the comparison.

9.1 What is already known

Almost symmetric laws, a world with a direction. Microscopic laws do not distinguish the past from the future. There is only a small exception, in weak interactions. Weak interactions are one of the forces of nature, the one of certain radioactive decays. The exception was measured directly with particles called B mesons (BaBar 2012). Where, then, does the direction we see come from? According to the most widespread idea, it comes from an initial condition. The universe started in a very ordered state. Since then entropy has grown (Lebowitz 1996). Entropy is a measure of disorder.

Heat can go backwards. Take two bodies. Each starts at its own temperature. And they start *without correlations* between them. They exchange energy only with each other, in a process that conserves total energy and entropy. Then they exchange heat only from hot to cold.

Now suppose instead that they start correlated. Correlated means that the properties of one are not independent of those of the other. Two dice loaded so as to always give the same number are correlated. Then heat can pass from cold to hot. But only by spending correlation, that is, by using up a little of their link. There is a precise rule. One takes the heat received by each body and divides it by the temperature of that body. Then one adds the two results. This sum cannot fall below the change of the mutual information of the two bodies. And so a backward passage requires the mutual information to decrease (Partovi 2008; Jennings and Rudolph 2010).

Mutual information says how much knowing the state of one body tells about the state of the other. It is measured in bits. A bit is the information of a yes-or-no answer.

The measurements.

- Micadei and others (2019) observed backward heat with two nuclear spins of a molecule. A nuclear spin is the spin of the nucleus of an atom. Heat passed from the colder to the hotter for about two thousandths of a second. Then it stopped, when the initial correlation had been spent.
- Naghdi and others (2022) obtained it in a chain of three qubits on a quantum computer. A qubit is the bit of quantum computers: it can be in 0, in 1 or in a superposition of the two.
- Huang and others (2026) obtained it in a system of four spins. There, for the same correlation, the direction of heat depends on the phase of the coherence. Coherence is the phase link between the parts of a superposition.

The direction of heat is not the whole direction of time.

- Latune and others (2019) show cases in which heat goes backwards, while entropy production stays positive. For them the arrow of time is not reversed.
- Jennings and Rudolph (2010, in their comment on a work by Maccone) give an example in which local entropy decreases while memory grows.
- For Hartle (2005), in the ideal model a record forms without a necessary increase of total entropy. It is erasure that requires it.
- Wolpert and Kipper (2023) study the memories they call Type 3. They are memories that start from a "ready" state. They need an initialisation, that is, a preparation, that increases global entropy.

Complexity, growth, point of view.

- The complexity of a quantum state is how many elementary operations are needed to prepare it. It grows for a long time like an entropy (Brown and Susskind 2018). Its maximum rate is fixed by the energy (Brown and others 2016). This result starts from the Margolus–Levitin bound of section 1.
- In causal sets spacetime is made of separate elements. There the passage of time is the birth of new elements (Rideout and Sorkin 2000; Dowker 2014).
- Rovelli (2015) proposes that the arrow of time depends on how whoever observes groups the states.
- Caticha (2011) builds an "entropic" time. It is made of inference steps, that is, steps of reasoning from the data. This time has a built-in direction.

In short: known physics already distinguishes the direction of heat from other arrows. There are the arrow of memory, that of complexity, that of growth. And known physics ties the direction of heat to the correlations between the observed parts.

9.2 How we tested the author's sentence

We translated the sentence «aumento del numero di relazioni legate a rete» ("increase in the number of relations linked in a network") in twelve ways. In addition there is a reference, known physics. All were registered before the calculations. Here are some examples of the twelve translations:

- the number of pairs of parts correlated above a threshold;
- the total correlation of the network, counted in bits;
- the sum of the mutual informations of the pairs;
- the relations along direct links only;
- the redundancy of the copies, that is, how many times the same information is repeated;
- complexity;
- the growth of the row of events;
- the ratio between two rhythms of time;
- the loop of deposit 2.48;
- the number of readings, compared with the information carried by each.

Then we compared them with three test benches. A test bench, here, is a numerical model reconstructed from published experiments and papers. The calculation is exact and has no fitted parameters, that is, numbers chosen on purpose to make the calculations come out right.

- **The two spins of Micadei.** The program recovers the backward heat up to 2.199 ms (milliseconds). The published paper gives about 2.1 ms.
- **The three-qubit chain of Naghdi.** The program recovers it in its qualitative features, that is, in its general behaviour.
- **The three qubits of Jennings and Rudolph.** Here the program does not recover one point of the published example: the one in which the coldest part gives up heat. It recovers it if the relative phase between the initial correlation and the interactions is inverted. But that is a physically different model. The reason for the difference with the published paper was not found. The counts of this bench hold for the registered reconstruction. The main result of §9.3 does not depend on them. Indeed, the first kind of opposite moments occurs in the benches of Micadei and Naghdi. The second kind occurs in that of Naghdi.

A correction, stated at once. In the first test we had added a criterion of our own. It said: a quantity reads the direction of time only if it decreases every time heat goes backwards. This link between direction and heat is not in the author's sentence. And nobody has tested it. The author observed that the direction of heat could, in some cases, be contrary to the complication of the matrix.

The criterion was removed. Two readings were "excluded" by that criterion. They were the number of pairs above a threshold and the relations along direct links only. Now they are no longer excluded.

There was also a second problem. The first test counted the relations between the observed parts. It did not count those inside the matrix of whoever observes. For this reason we registered a second test, with an **observer's record**. The record is made of a few memory cells. The cells start empty, with no energy of their own. Then they copy the energy level of an observed part.

The **complication** is the number of relations that the record carries with the observed parts. It is counted in bits. In technical terms it is the mutual information between the record and the parts. With several cells the correlation among the memories is also measured, separately. This does not enter the comparisons with the direction of heat. In two benches we took one of the parts itself as the observer's memory.

9.3 What comes out

The relations between the observed parts follow the heat. Let us count the correlations between the observed parts in bits, over the whole network. This is the **total correlation**. It decreases every time heat goes backwards. It is an exact identity. The program checks it to within $5.2 \cdot 10^{-14}$. It is the Jennings–Rudolph bound extended to several parts.

The converse does not hold. Correlation can also decrease when heat goes in the normal direction. This is known physics. And it concerns the observed parts, not the matrix of whoever observes.

The sum of the mutual informations of the pairs also decreases at every reversal of heat. This holds in the seven cases tested.

Two readings, instead, do not follow the heat: the number of correlated pairs above a threshold and the relations along

direct links only.

- For every threshold tested there is a stretch in which heat goes backwards and the number of pairs does not decrease.
- In the Jennings–Rudolph bench the relations along direct links stay at zero while heat goes backwards.

These two readings would need a threshold fixed by an independent criterion. Today this criterion does not exist.

The relations in the observer's record do not always go with the heat. In every bench tested there are moments with opposite signs with respect to the initial values. They are of two kinds:

- backward heat, and complication above its initial value;
- or heat in the normal direction, and complication below its initial value.

Here are the cases one by one.

- **Two spins of Micadei, one reading.** Heat goes backwards for 2198 instants. In all these instants the record, which started empty, carries between 0.365 and 1.166 bits of relations. And the reading does not change the heat. It is almost obvious by construction. It only says that nothing prevents the record from growing while heat goes backwards.
- **Four readings during the process.** The record rises at each reading: 0.73, then 1.04, then 1.24, then 1.38 bits. The readings reduce the amount of backward heat. At 1.0 ms the backward heat is −0.41 peV, against −0.71 without readings. At 1.5 ms it is −0.30, against −0.61. The peV, or picoelectronvolt, is a millionth of a millionth of an electronvolt. But the readings make the backward heat last until the end of the computed time. We had registered an expectation: that after the first reading the backward heat would stop. It did not go that way.
- **Three qubits of Jennings and Rudolph, with one of the parts as memory.** Here the observer's memory is itself the correlation that heat spends. There are 1366 points in which heat goes backwards. In all of them the complication is below its initial value, which is 0.520 bits. We had registered an expectation: at least one point with backward heat and complication above the initial value. This expectation did not come true. There are, however, 12,754 points of the other kind: heat in the normal direction and complication below the initial value. Another 125 points are not classified. Their changes are within the tolerance, that is, they are so small that they count as zero.
- **Chain of Naghdi, with one of the parts as memory.** In one case there are 60 instants with backward heat and complication above the initial value. In another case there are none. But there are 931 instants with normal heat and complication below. In each case one instant is not classified, because it is within the tolerance.

What seems to decide. When the observer writes new records, its complication rises, whatever the direction of heat. When its memory is the very correlation that heat spends, the complication is below the initial value every time heat goes backwards.

The author's sentence says that the direction of heat can be, in some cases, contrary to the complication of the matrix. This sentence therefore has examples in standard quantum mechanics. It holds for these models and for this way of counting. It does not hold as a general law.

The number gives a bound, the phase chooses the direction. Take two initial states of the two spins of Micadei. They have the same temperatures, the same mutual information and the same counts. Yet they give heat in opposite directions in the first 0.2 ms. No quantity of number or amount, by itself, fixes the local direction of heat. It is chosen by the phase of the correlation. It happens as in the experiment of Huang and others.

The other readings.

- Complexity is not measured today. So it is not testable.
- The growth of the row of events is compatible as a reading of the passage of time.
- The number of readings grows by one at each reading. The information carried by one reading, instead, falls as 1/T, that is, one divided by T: if T doubles, that information halves. The program recovers the known asymptotic value to within 0.01 %. Asymptotic means the value that one approaches after very many readings. The "information time" grows as the logarithm of the number of readings. The logarithm grows more and more slowly: to make it rise by one step, the number of readings must be multiplied, not just increased. This is a known result of Bayesian statistics, the statistics that updates the estimates with each new datum. It has the precedent of Caticha.

Here are all the readings and their outcomes.

reading	outcome
known physics (reference); coarse-grained entropy	compatible
correlated pairs above a threshold	does not follow the heat; not conforming until the threshold is fixed by an independent criterion
total correlation between the observed parts, in bits	follows the heat (exact identity); compatible, not distinctive
sum of the mutual informations of the pairs	follows the heat in the cases tested; compatible, not distinctive
relations along direct links only	does not follow the heat; not conforming until the threshold is fixed by an independent criterion
redundancy of the copies	compatible, not distinctive
complexity	not testable today
growth of the row of events	compatible as the passage of time
two rhythms of time at the same point	compatible if the ratio is the same for every process; not verified directly
the loop (deposit 2.48)	compatible, not distinctive
the observer's network (the part of the network that contains the record)	compatible, not distinctive
number of readings and information per reading	compatible, not distinctive (precedent: Caticha 2011)
complication of the observer's record	**compatible, not distinctive; it does not always go with the heat: in all the benches tested there are moments with opposite signs with respect to the initial values**

Coarse-grained entropy is the entropy calculated by looking at things from far away, without the details, as one looks at a grainy photo.

No reading gives today a prediction different from known physics. Read as the complication of the observer's record, the direction of time is a quantity different from the direction of heat. At some moments it agrees with the heat, at others it is opposite.

9.4 The contact with mass

In §4.3 we saw that a mass at rest, in a state of definite energy, produces no distinct states. Its state changes only by a phase. As long as it stays isolated, therefore, its complexity does not grow. And it brings no new relations with a record. When it interacts with a record, instead, it can become correlated with it, like any system.

One could read this as follows: "the contained multiple does not contribute to the direction of time, which arises when something is read". This reading is an interpretive hypothesis of the AI. The benches of this section do not test it.

The maximum rate at which complexity can grow is fixed by the energy. It passes through the same Margolus–Levitin bound that in §4.3 is the bridge between mass and time. This is a consistency between the two parts of the work, not a proof.

There is a difference with the eleven translations of mass of section 3. The direction of time was translated using the observer's record. Like item M-i of section 10.

9.5 What is not claimed

- The direction of time is not derived. It is read. And the reading is compatible with known physics.
- It is not claimed that the complication of the record is always contrary to the heat. Nor that it usually is. At some moments the signs agree, at others they are opposite.

- The observer is a minimal model and the benches are small. The complication is counted in bits. Other ways of counting, for example complexity, were not tested.
- Two registered expectations did not come true. One validation was not reproduced. And one registered control formula was written wrongly; the right one holds exactly. Everything is reported in the outcome notes, that is, in the documents that describe the outcomes (see at the end).

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10. Mass in the matrix of whoever observes (item M-i)

New section in version 2.64.1. The eleven translations of section 3 read "contained" as a rhythm closed inside the observed body. The author instead speaks of a multiple «contenuto direttamente nella matrice del punto di osservazione» ("contained directly in the matrix of the point of observation"). This section writes and tests the translation that was missing. An external review pointed it out.

The translation

The observer copies the same body at each tick of its own clock. It is like taking many photos of the same object, one at each tick-tock. The row of copies divides into two parts.

- **What changes from one copy to the next.** It can be seen only by comparing two copies. It is the read multiple, that is, time.
- **What stays the same in all the copies.** It can be read from a single copy. It is the contained multiple.

Mass is the size of this second part. Its value is the rhythm that the observer's time gives to a contained block. What stays the same divides into blocks, that is, into separate groups. The value exists only if the block has a single rhythm. In technical terms: if the block is an eigenspace of the generator of time. This clarification was written after the first calculations.

Now let us think of observers who move in different ways. What do they have in common? Here *in common* has the sense of the common part of deposit 2.62 (Gács and Körner 1973). What they have in common is mass as we usually understand it.

In the model the ticks follow the ordinary time of quantum mechanics. No internal clock is built. Protocol, programs and notes were registered before the calculations. A protocol is the written plan of the test. Then they were verified by the second artificial intelligence, in two rounds. The choices that tie the author's words to the calculated quantities are the AI's. They are listed in the outcome note.

1. What is conserved stays contained

Take a test body made of two bound pieces.

- The copies of the total energy stay identical, at any distance between the ticks. Everything is contained: 1.747 bits.
- The copies of the levels of the single pieces keep only the number of excitations: 1.271 bits. An excitation is a packet of extra energy with respect to the lowest state. The number of excitations is a count. It too is contained.
- The copies of one piece alone keep nothing.

This is what is known from non-demolition measurements (Braginsky, Vorontsov and Thorne 1980). A non-demolition measurement can be repeated and gives the same outcome again. The quantities that are conserved give the same outcome again.

2. The value weighs, binding included

The bound block has value 0.7197, in model units. The mean count of the pieces is 0.8445. The value is lower by 0.1248. This difference is the binding energy, given by the model formula g^2/Ω, in which g measures the strength of the binding and Ω is another quantity of the model, a frequency that depends on g and on the energies involved. It is like a nucleus, which weighs less than its pieces.

The count is as contained as the value. But the count does not have a single rhythm. What selects the value is the rule that gives a value only to blocks with a single rhythm. This rule is a choice of the AI.

There is also the opposite choice: counting the copies of the pieces. With this choice the item does not follow from the data. If it were adopted, it would give the numbers that §4.2 excludes: for MICROSCOPE, $2.7 \cdot 10^{11}$ standard deviations.

What stays contained, and which blocks receive a value, depends on two things: on what the observer copies and on how it ticks.

- If it copies one piece alone, no block receives a value.
- If it ticks at a fixed rate, that is, in a stroboscopic way, more seems contained. It happens as with flashing lights in a disco, which make spinning things look still. But the additional blocks receive no value.

3. What observers in motion have in common distinguishes the masses

Two observers with different velocities record the energy and momentum of five particles. For a slow body momentum is the mass times the velocity; for very fast particles the formula changes, and even light, which has no mass, carries momentum. The particles are:

- the electron;
- the muon, a particle similar to the electron but heavier;
- the proton;
- the neutron;
- the deuteron, the nucleus of heavy hydrogen, made of a proton and a neutron.

Each record notes the values in cells. A cell is like a square on a sheet of squared paper: one notes only which square a value falls in. The smaller the cell, the more precise the record. The cells are measured in MeV, that is, in millions of electronvolts.

- With cells of 0.01 and 0.1 MeV the common part of the two records separates the five particles. Each common component contains the invariant mass. For the proton, with cells of 0.01 MeV, the interval runs from 938.245 to 938.299 MeV.
- With cells of 1 and 10 MeV proton and neutron merge.
- With energy alone they merge at every resolution, that is, at any size of the cells.

The energy of a photon, that is, of a grain of light, is not common to the observers. And its invariant is zero. Two opposite photons closed in a box, instead, give an invariant of 2 MeV.

This is mass as the invariant of relativity (Wigner 1939). Invariant means that all observers find the same value. Here it is read as the common part of the records. This reading is the AI's, not Wigner's.

4. Complication and mass of the record are different quantities

Let us take again the record of section 9, with cells that have no energy of their own. The four readings carry 0.731, 1.044, 1.237 and 1.376 bits of relations. And they carry no energy.

Now let us give the cells an energy gap: $h \cdot 1\ \text{kHz}$. In words: Planck's constant times a thousand cycles per second. With the same relations, the energy of the record rises to 2.75 peV. Four excited cells that read nothing carry 16.5 peV and zero bits.

In the model the mass of the record follows its energy, not its complication.

5. In no case examined does the item give a number different from standard physics

- **Clocks in gravity.** The item coincides with M-e, that is, with general relativity.
- **Observers in motion.** It coincides with special relativity.
- **Atoms in a superposition of two internal states.** The result of Rosi and others (2017) is compatible with zero at 1.1 standard deviations. The item does not give, for this comparison, a prediction different from the standard one.
- **No suitable quantity.** It can happen that the observer copies no conserved quantity with a single rhythm. Then the item gives no value. It is not conforming, not distinctive.

There is a nearby reading, proposed by Vopson (2019). It says that mass is information multiplied by $k_B T \ln 2 / c^2$. Here k_B is Boltzmann's constant, which ties energy and temperature. T is the temperature. $\ln 2$ is the natural logarithm of 2, a fixed number. This reading has three problems.

- It introduces a constant and a temperature: it does not respect the criterion on new constants.
- In the record without an energy gap it would give a mass without energy. So it does not follow the criterion $E = mc^2$. This depends on the choice that ties bits to mass.
- Today it is not testable in the cases of the protocol.

The quantum phase of free fall in gravity was not compared.

Here are the five forms of item M-i and their outcomes.

item	what it says	outcome
M-i1	mass = the part of the copies that stays the same at every tick; value = rhythm, only for blocks with a single rhythm; common to observers in motion	compatible, equal to standard physics by construction; not conforming where no copied block has a single rhythm (for example copying one piece alone)
M-i2	as M-i1, with value E/c^2 of the single observer	does not follow the limiting case of light (depends on a choice of the AI)
M-i3	as M-i1, with value = contained copies of the pieces	does not follow from the data (MICROSCOPE, as M-d1; depends on a choice of the AI)
M-i4	mass = energy of the record	compatible as an identity of definition in the model
M-i5	(an item of the AI) mass = complication in bits × $k_B T \ln 2 / c^2$ (Vopson 2019)	does not respect the criterion on new constants; $E = mc^2$ does not follow in the record without an energy gap (depends on the choice that ties bits to mass); not testable today in the cases tested

In simple words:

- M-i1 is compatible, equal to standard physics by construction, but not conforming where no copied block has a single rhythm;
- M-i4 is compatible as an identity of definition;
- M-i2 does not follow the limiting case of light, and M-i3 does not follow from the data, like M-d1: these two outcomes depend on choices of the AI;
- M-i5, an item of the AI, does not respect the criterion on new constants. In addition $E = mc^2$ does not follow in the record without an energy gap (it depends on the choice that ties bits to mass). Today it is not testable in the cases tested.

What it adds

This item translates the author's words: the matrix of whoever observes, and not a property of the body. And it coincides with $E = mc^2$ and with relativity, within the measurements used in section 4.

The author distinguishes between read multiple and contained multiple. In the record this distinction corresponds to the division of the copies: what changes and what stays. It is a consistency with two things:

- with non-demolition measurements;
- with the framework of Page and Wootters (1983), in which time arises from correlations with a clock. This framework was illustrated in the laboratory by Moreva and others (2014) and developed in theory by Giovannetti, Lloyd and Maccone (2015).

The model does not test that mechanism.

The common part of deposit 2.62 can be applied in two ways. Applied to the copies of a single observer in time, it gives what is conserved. Applied to observers in motion, it distinguishes the masses.

There is prior work to be cited first.

- De Broglie (1924), who associated with mass a periodic phenomenon, with $h\nu = mc^2$.
- Noether (1918), for the link between energy and symmetry in time.
- Wigner (1939), for mass as an invariant.

What is not claimed

- It is not claimed that MM derives mass.
- It is not claimed that mass depends on the observer. Two things depend on the observer: what stays contained in its copies, and the energy in its frame, as in relativity.
- It is not claimed that the value does not depend on how one copies. Where the value is assigned it is, by construction of the rule, the rhythm of the block.
- No new prediction is claimed.

How the work was done

Who did what. The author is not a physicist. The translations, the programs and the calculations were made with an artificial intelligence (Claude, Anthropic). A second artificial intelligence (gpt-6-luna via Codex, OpenAI) verified them, in two rounds. In the list of the package the verifications carry the name Astra.

- In the first round ten remarks arrived: eight accepted, one accepted with a different argument, one accepted in part.
- In the second round the remaining remarks were accepted, except one. It concerned the Brannen reference, which is not a journal article but a conference talk.
- Thanks to these verifications, two results were corrected: the copies of the Compton rhythm (from excluded to not concluded) and the whole body on the checkerboard (from excluded to not conforming).
- Three external reviews of the text then improved its clarity.
- A verification of the package by the second artificial intelligence corrected some sentences and the program that redoes the calculations.

The author's sentences from which the work arises are in the background file, AUTHOR_BACKGROUND.md.

Version 2.64.1. An independent check requested by the author was done by another instance of Claude. This instance did not know the conclusions of the first. It found the undeclared translation choices, corrected in this version. Each point was rechecked against the original files.

- The corrections and section 9 were then verified by the second artificial intelligence. The reports and the answers are in the folder 04_controllo_262_264. The details of the corrections are in 04_controllo_262_264/PIANO_CORREZIONI_v1.md.
- An external review of the corrected version then added two clarifications, in §4.4 and in §9.2.
- Item M-i of section 10 was suggested by the same review. It was verified by the second artificial intelligence in two rounds. The work on item M-i is in the folder 06_massa_matrice_osservatore.
- The work on time of section 9 is in the folder 03_direzione_tempo. It was verified by the second artificial intelligence in four rounds: two on the first test and two on the test with the observer's record. The outcome notes cited in §9.5 are also there.

The registrations. Every protocol and every program has a SHA-256 fingerprint and a UTC time, registered before the calculation was started. The results are in JSON files. The rules for neutrinos (§4.9) were registered on 24 October 2026.

How to reproduce: mass (sections 1-4). Three programs in Python 3 are needed, with the sympy library. They run in a few seconds in all on an ordinary computer.

- livello1_massa.py: the registered calculation.
- correzione_1.py: the numbers asked for by the first verification.
- numeri_del_testo.py: the examples of §1 and §2.

The AME2020 atomic-mass file (mass_1.mas20.txt) and the Particle Data Group constants are needed. The protocol, the two corrections and the two independent verifications are in the package.

How to reproduce: time with a direction (section 9). Five more programs in Python 3 are needed, with the numpy and scipy libraries. They run in about three minutes in all.

- livello1_tempo.py: the registered calculation of the first test.
- diagnosi_V4_JR.py and controlli_giro1.py: the checks after the calculation and after the first verification.
- livello1b_registro.py: the test with the observer's record.
- controlli_1b_giro1.py: its checks.

They are in the folder 03_direzione_tempo, with the protocols, the outcome notes, the corrections and the four independent verifications.

How to reproduce: mass in the matrix of whoever observes (section 10). Two more programs in Python 3 are needed, with the numpy library. They run in a few seconds: mi_massa_osservatore.py and correzione_1_mi.py. They are in the folder 06_massa_matrice_osservatore, with the protocol, the outcome notes, the corrections and the two independent verifications.

Package contents. The contents are described in README.md, which is the same text in English (sections "Package contents", "How to run", "Related records"). In short, the package contains:

- the registered protocol, the two corrections, the outcome notes;
- the three programs with the results;
- the two verifications by Astra on the calculations and the one on the package;
- the folder 03_direzione_tempo of section 9: protocols, outcome notes, corrections, five programs with the results, four verifications by Astra;
- the folder 06_massa_matrice_osservatore of section 10: protocol, outcome notes, corrections, two programs with the results, two verifications by Astra;
- the plan of the corrections of version 2.64.1 and the verifications by Astra on it (04_controllo_262_264);
- the registers with the fingerprints;
- run_all.py, which redoes all the calculations and compares them with the registered ones.

The command is `python run_all.py mass_1.mas20.txt`. It takes about three minutes. The program ends with "ALL MATCH".

The AME2020 atomic-mass table is not redistributed. It can be downloaded from the IAEA (www-nds.iaea.org/amdc/ame2020/mass_1.mas20.txt). The program run_all.py checks its fingerprint. The author's sentences are in AUTHOR_BACKGROUND.md. The licences are in LICENSE.md: texts CC BY 4.0, code MIT.

Related records.

- The twin deposit on observed objects, 2.62, corrected in version 2.62.1.
- Deposit 2.48 on the loop (DOI 10.5281/zenodo.23352814).
- Deposit 2.25, in record 2.27 (DOI 10.5281/zenodo.22667516).
- 2.36.2 (DOI 10.5281/zenodo.23287965).
- 2.38 (DOI 10.5281/zenodo.23055978).
- 2.40 (DOI 10.5281/zenodo.23280168).

This version. This student version was written with an AI assistant (Claude) from the archived Italian text (LEGGIMI_IT.md). The numbers were checked with a program against the archived text.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **Backward heat:** the passage of heat from the colder body to the hotter one. It is possible only if the bodies start correlated. And it is limited by the correlation that is spent.
- **Binding energy:** the energy that must be given to a body to separate its pieces.
- **Bit:** the unit of information: that of a yes-or-no answer.
- **Causal set:** a model in which spacetime is made of separate elements, linked by relations of before and after. The passage of time is the birth of new elements.
- **Complexity (of a quantum state):** the minimum number of elementary operations needed to prepare it.
- **Complication (of the observer's record):** in §9, the number of relations, counted in bits, that the record of whoever observes carries with the observed parts. With several cells the correlation among the memories is also measured, separately.
- **Compton frequency:** the rhythm of a mass, $\nu = mc^2/h$.
- **Contained (contained multiple):** for the author, the multiple contained directly in the matrix of the point of observation. In the AI's reading used in this work, the counting that stays inside the thing as a rhythm, without producing distinct states: it is mass. In §10 (item M-i), the part of the observer's copies that stays the same at every tick.
- **Copy:** in the author's framework, a repetition of the common unit. In experiments, a recorded mark that carries the imprint of something. The mark can be in an internal degree of freedom of the particle or in the environment,

for example in a scattered photon. It can carry the value of an object (as in deposit 2.62) or the path taken (as in §4.4).

- **Correlation:** the link between two parts, when the properties of one are not independent of those of the other.
- **Count of the pieces:** the sum of the masses of the pieces of a body taken one by one, without the binding.
- **Degree of freedom:** a way in which a system can change, for example its position or its spin.
- **Deposit:** a work of the author archived on Zenodo, with a number such as 2.40 or 2.62.
- **Distinct (orthogonal) states:** states that a measurement can distinguish without error.
- **Eigenstate (of energy):** a state with definite energy. In time it changes only by a phase, while nothing observable changes.
- **Electronvolt (eV):** a very small unit of energy, used for particles. An MeV is a million eV, a GeV a billion, a meV a thousandth, a peV a millionth of a millionth.
- **Entropy:** a measure of disorder, or of the information missing to know the state. In §9 it is counted in bits.
- **Equivalence principle:** all bodies fall in the same way, whatever their composition.
- **Feldman–Cousins (method):** a way of computing a limit that takes into account that a quantity cannot be negative.
- **Fractions:** in the author's framework, the copies arranged side by side, all together. They appear as space.
- **Fringes (contrast of):** the light and dark stripes of interference. The contrast says how sharp they are.
- **GCD (greatest common divisor):** in the author's framework, the unit that two entities have in common and that is counted in each.
- **Gravitational redshift:** the slowing of rhythms near a mass, seen from far away.
- **Interferometer:** an instrument that splits a particle (or a wave) into two paths and recombines them, to see the fringes.
- **Invariant mass:** the mass that observers in any motion find equal. It is obtained from energy and momentum. In §10 it is what their records have in common.
- **Item:** a row of the list of tests, with a short name such as M-a or M-d1. Each item is a translation, a reference or a threshold.
- **Komar (mass):** the mass of a body read from the way it slows the clocks around it, summed over a surface that encloses it.
- **Lunar laser ranging:** the measurement of the distance of the Moon with laser pulses reflected by mirrors left on the Moon.
- **Margolus–Levitin (bound):** the maximum rate at which a system of given energy can pass through distinct states.
- **Mass at rest, inertial, gravitational:** the mass at rest is the energy of the body at rest. Inertial mass is the resistance to a push. Gravitational mass is what is attracted (passive) or what attracts (active). The equivalence principle says that inertial mass and passive gravitational mass are the same. So all bodies fall in the same way. The equality with active mass is checked separately, for example with lunar laser ranging.
- **Mass defect:** the fact that a bound body weighs less than its separate pieces.
- **Matrix:** in the author's framework, the set of the copies of the common unit, counted from a point of view.
- **MICROSCOPE:** a satellite that compared in orbit the fall of two cylinders of titanium and of platinum.
- **Multiple:** in the author's framework, counting the common unit forward, one copy after another.
- **Mutual information:** how much knowing the state of one part tells about the state of another, counted in bits.
- **Neutrino ordering:** which of the three neutrinos is the lightest. It is normal if the lightest is the first.
- **Non-demolition measurement:** a measurement that, when repeated, gives the same outcome again, because the measured quantity is conserved.
- **Nordtvedt (effect):** the hypothesis that bodies with much gravitational binding energy fall in a different way. Lunar laser ranging excludes it.
- **Observer's record:** in §9, a few memory cells that start empty and copy the state of an observed part. It is the AI's translation of the matrix of whoever observes.
- **Outcome:** the result of the comparison of an item with the data: excluded, compatible, not testable today, distinctive, not conforming, not concluded.
- **Parametrisation:** a precise way of writing an idea as a formula, so that it can be compared with measurements.
- **Phase:** the angle of a rhythm, like the position of a clock hand. A phase that changes in the same way across the whole state is not observable on its own.
- **Physical time:** the time measured by clocks. In the thesis it is the read multiple.
- **Planck (mass, time):** the scales built with h, c and G: about $2.2 \cdot 10^{-8}$ kg and $5.4 \cdot 10^{-44}$ s.
- **Premise:** a starting sentence of the author, like a postulate. The exact words are in the background file.
- **Proper time:** the time marked by a clock that travels with the body.
- **Qubit:** the bit of quantum computers. It can be in 0, in 1 or in a superposition of the two.
- **Read (read multiple):** in the thesis, the counting that becomes a row of distinct states. It is time.
- **Reading:** in this work, a precise way of understanding a sentence of the author (a translation), or the way in which an observer records something. A negative outcome concerns the reading tested.
- **Record:** a memory, the place where the marks of what has been observed remain.
- **Registered:** written in a file whose fingerprint and time were noted before the calculation. The file is then not changed.

- **Spin:** a quantum property of particles and nuclei that resembles a small rotation on themselves, but it is not a spinning ball.
- **Standard deviation (σ):** the measure of the uncertainty of a datum. A result three standard deviations away from a prediction casts strong doubt on it. At 10^{11} standard deviations it excludes it.
- **Step (of the checkerboard):** the interval of time and space between two squares.
- **Superposition:** a state that combines two conditions, each with its own amplitude (a number from which the probability comes); a measurement finds only one of them. The overlap of §4.3 is something else: how much two states resemble each other.
- **Test bench:** in §9, a numerical model reconstructed from an experiment or from a published paper.
- **Time with a direction:** the direction of time. We remember the past and not the future, and heat passes from hot to cold.
- **Total correlation:** the sum of the entropies of the parts, minus the entropy of the whole. It says how many relations, in bits, link the parts of a network.
- **Translation:** in this work, a mathematical definition of the thesis that allows a calculation comparable with the data.
- **Value:** in the thesis, the total energy contained in a thing, binding included. Divided by c^2 it is its mass.
- **Walker:** on the checkerboard, the particle that moves forward in a zigzag along the directions of light rays.

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